Julian Bartlett

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GitHub | LinkedIn | Portfolio

Professional Summary

Computer Science student seeking a Summer 2027 software engineering internship, with experience in backend systems, full-stack applications, and AI/computer vision. Built concurrent job processing, durable event ingestion, and an x86_64 hobby operating system.

Education

North Carolina A&T State University — Expected December 2027
Computer Science | GPA: 3.5

College of The Albemarle — June 2025
Associate of Arts | GPA: 4.0

Experience

Data Analyst Intern | AI / Computer Vision Project — May 2026 – Aug 2026
Texas A&M Transportation Institute (TTI) — *College Station, TX*

- Researched computer-vision methods for detecting people in smoke and fire, supporting an AI-enabled drone workflow for emergency response.
- Evaluated detection challenges in low-visibility environments and translated findings into system requirements and development priorities for a computer-vision pipeline.

Mobile Expert — Mar 2025 – Aug 2025
T-Mobile — *Kitty Hawk, NC*

- Resolved device, activation, and service issues through structured troubleshooting; explained technical solutions in clear, actionable steps for customers.

Projects

TaskForge – Distributed Job Queue | *Python, FastAPI, PostgreSQL, Redis, Docker, pytest* — Source

- Built a task-processing system with PostgreSQL job history, Redis priority queues, concurrent workers, fenced leases, retries, and execution timeouts.
- Verified 100 jobs across 8 workers and recovery of all 10 deliberately injected transient failures; containerized the application with Docker and tested with pytest.

SignalDock – Cloud Log & Analytics | *Python, FastAPI, PostgreSQL, Redis Streams* — Source

- Built durable event ingestion with Redis Streams consumers and replay-safe PostgreSQL indexing for searchable logs, error-rate and latency analytics, and worker monitoring.
- Added automated integration and Docker tests with GitHub Actions to validate ingestion and processing.

Bax OS / Hobby Operating System | *C, x86_64 Assembly, Zig* — Source

- Built an educational x86_64 operating system with a BIOS bootloader and freestanding C kernel; implemented VGA and serial output, interrupts, timer and keyboard support, and a kernel shell.

Drone Fire Intelligence | *Python, OpenCV, YOLOv8, scikit-learn, DVC* — Source

- Built a multi-model pipeline for normal and thermal drone footage to detect people, vehicles, fire, and thermal targets; classified operational risk, recorded detection evidence, and versioned the ML workflow with DVC.

Technical Skills

Languages: Python, Java, C, TypeScript, HTML/CSS
Backend: FastAPI, Node.js, PostgreSQL, Redis, Redis Streams, REST APIs, WebSockets
Cloud / DevOps: AWS, Docker, GitHub Actions, Vercel
AI / ML: OpenCV, YOLOv8, scikit-learn, DVC | **Frontend:** React, Next.js
Systems / Tools: Linux, x86_64, Git, GitHub, pytest